



EdgeSafe

An Edge-Based Weapon Detection and Secure Event
Logging System

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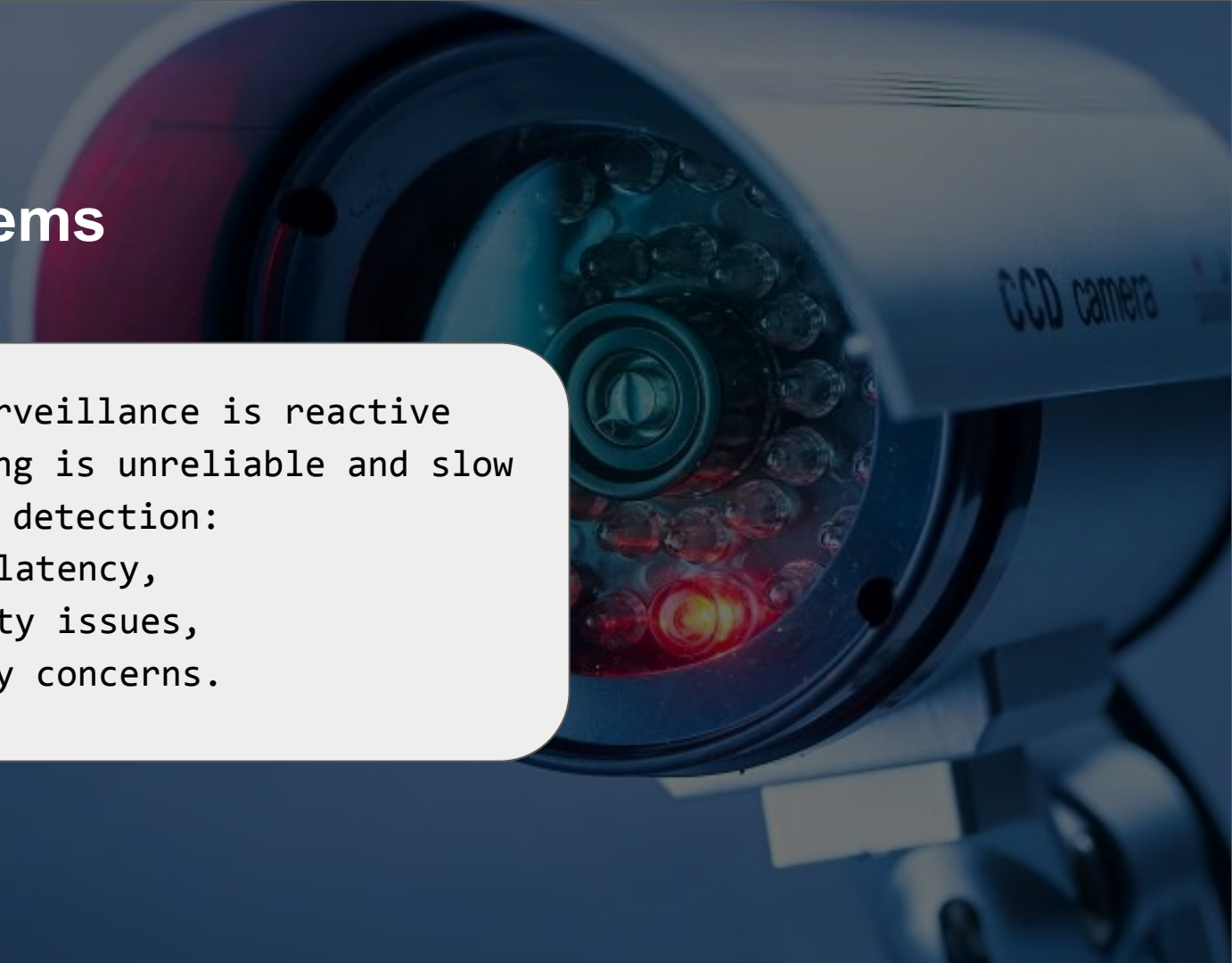
Motivation





Current Problems

- Traditional surveillance is reactive
- Human monitoring is unreliable and slow
- Cloud-based AI detection:
 - introduce latency,
 - connectivity issues,
 - and privacy concerns.





Why It Matters

- Firearms are now a leading cause of death among youth in the U.S.
- Faster detection can improve situational awareness and response time

EdgeSafe explores local detection as a faster and a more controlled alternative.



Objectives





Specific Objectives:

- **Detection**
 - $\geq 80\%$ detection accuracy
 - ≤ 1 second inference latency
- **Automation**
 - Autonomous detection pipeline
 - Real-time result publishing
- **Security**
 - HTTPS communication
 - Authenticated dashboard access
- **Auditability**
 - Structured event logging
 - Historical event retrieval



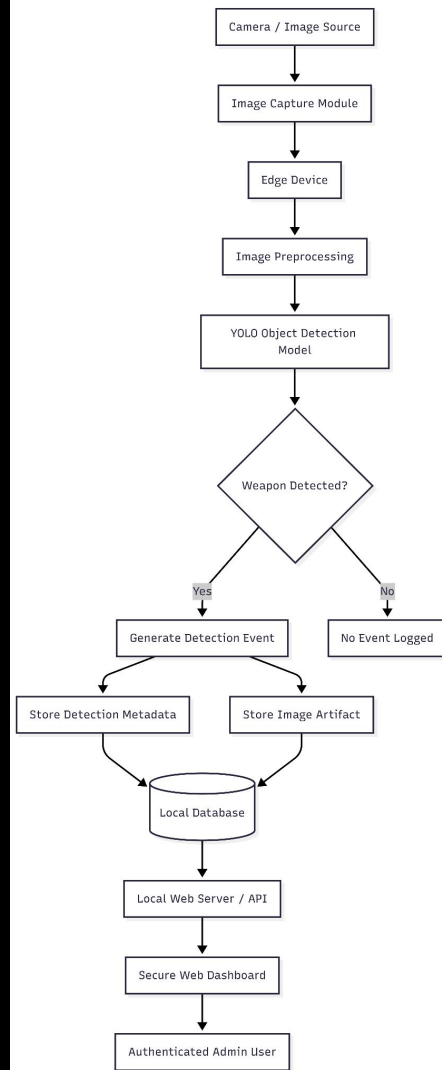
Solution Approach





System Overview

1. Image input
2. Edge inference
3. Weapon detection
4. Event generation
5. Database logging
6. Dashboard visualization





System Architecture

Layout:

Presentation Layer

- React/Vite dashboard

Application Layer

- FastAPI backend
- YOLO inference
- Authentication

Data Layer

- SQLite event logging
- Image references

- NVIDIA Jetson Nano



Detection Model

- YOLOv11 object-detection architecture (Ultralytics), chosen for its balance of speed and accuracy on edge devices
- Transfer learning applied: pretrained weights fine-tuned on a firearm-specific dataset rather than training from scratch
- Two candidate variants evaluated for deployment: YOLOv11n (nano) and YOLOv11s (small)
- Models evaluated with standard object-detection metrics: precision, recall, F1-score, and mAP
- Confidence-threshold tuning applied under a $\geq 90\%$ precision constraint to minimize false alarms



Dataset used:
Kaggle [Gun Dataset YOLO v8](#)



Backend

- FastAPI-based REST API
- Structured JSON responses
- Event retrieval endpoints
- Error handling and validation

Event Logging

- SQLite local database
- Detection metadata storage:
 - timestamp
 - source
 - confidence
 - latency
 - image reference
- Secure local storage
- Traceability and auditability



Dashboard

- React (Vite) dashboard displays system status and detection of events.
- Event metadata display
- Empty-state handling prevents dashboard from breaking even when no events are logged.
- Dashboard retrieves event data directly from backend API endpoints.

Authentication

- JWT tokens are used to protect backend routes.
- Passwords are stored using bcrypt hashing.
- Login page restricts access to authorized users only.
- Authenticated sessions allow access to protected dashboard pages.



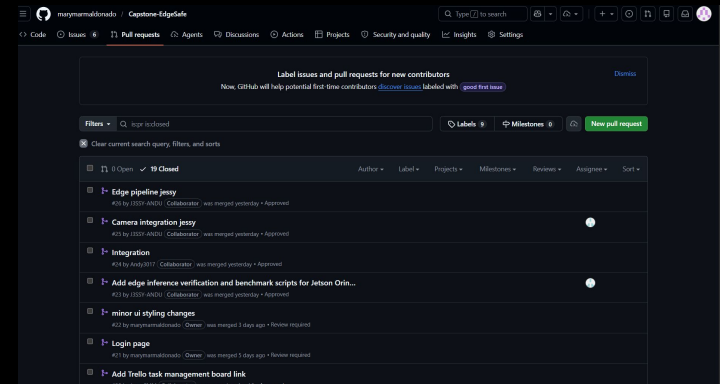
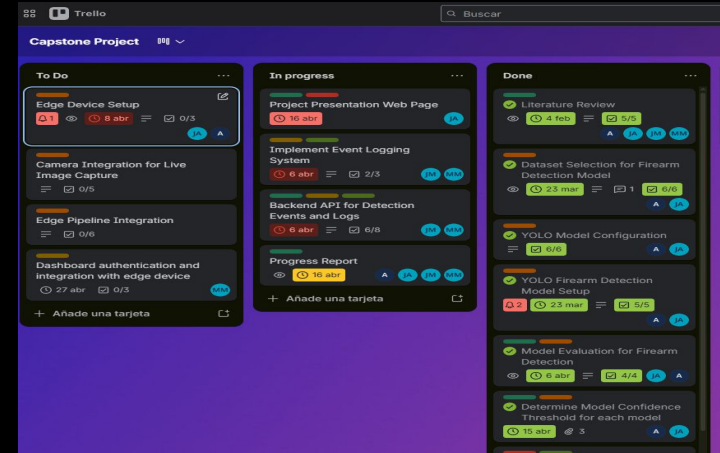
Results





Project Management and Documentation

- Trello Task Board was used to itemize issues and tasks.
- Git/GitHub was used to host repository and handle all source code.
- Feature branches and Pull Requests supported team collaboration and code review.
- Documentation was maintained throughout the project. Including: API documentation and project website containing project and member's information.





Backend, Database and API

Implemented Features:

- FastAPI REST API
- JSON event responses
- Detection event retrieval
- Error handling validation

```
JSON  Raw Data  Headers
Save Copy Collapse All Expand All Filter JSON
▼ [0...99]:
  ▼ 0:
    id: 1
    model_name: "YOLO"
    inference_ms: 12
    timestamp: "2026-04-15T16:50:48.346669"
    confidence: 0.6854119955376771
    detected: 1
    source: "CAM 4"
    image_path: "images/test0.jpg"
  ▶ 1: { id: 2, model_name: "YOLO", inference_ms: 9, ... }
  ▶ 2: { id: 3, model_name: "YOLO", inference_ms: 7, ... }
  ▶ 3: { id: 4, model_name: "YOLO", inference_ms: 7, ... }
  ▶ 4: { id: 5, model_name: "YOLO", inference_ms: 15, ... }
  ▶ 5: { id: 6, model_name: "YOLO", inference_ms: 5, ... }
  ▶ 6: { id: 7, model_name: "YOLO", inference_ms: 6, ... }
```



Dashboard and Authentication

- The dashboard displays logged detection of events clearly.
- The login page controls access to restricted dashboard areas.
- After login, the backend generates a JWT token for authenticated access to protected routes.
- This ensures only authorized users can access detection results and event logs.

The screenshot shows the 'EdgeSafe' dashboard with a navigation bar containing 'Home' and 'Results'. The main heading is 'Detection Results' with a subtitle 'Review recent firearm detection activity, detection metadata, and performance metrics.' Below this, there are three summary cards: 'Recent Events' showing 21 latest processed records, 'Average Inference' showing 10 ms recent processing latency, and 'Latest Detection' showing a detection on 4/15/2026 at 10:45:17 PM with 91% confidence. A table titled 'Recent Detection of Events' follows, listing three events with columns for ID, Timestamp, Source, Model, Inference, Confidence, Status, and Image Ref. All three events are marked as 'Detected'.

ID	Timestamp	Source	Model	Inference	Confidence	Status	Image Ref
#1	4/15/2026, 10:45:17 PM	CAM 1	YOLO	10 ms	91%	Detected	images/test1.jpg
#2	4/15/2026, 10:45:17 PM	CAM 2	YOLO	12 ms	87%	Detected	images/test2.jpg
#3	4/15/2026, 10:45:17 PM	CAM 3	YOLO	11 ms	76%	Detected	images/test4.jpg

The screenshot shows the 'EdgeSafe Login' form. It has a title 'EdgeSafe Login' and a subtitle 'Authorized access to detection dashboard'. Below the subtitle are two input fields: 'Username' and 'Password'. At the bottom of the form is a dark blue button labeled 'Sign In'.



Model Detection Performance

- Both exceeded the 80% target
- High precision yields a low false-positive rate, critical for maintaining operator trust
- YOLOv11s achieved slightly higher recall and F1; YOLOv11n achieved higher precision
- Both successfully detected and localized firearms in representative test imagery
- Results confirm transfer learning produced a reliable detector under project constraints

Table 2. Test Performance Under Precision Constraint of $\geq 90\%$

Model	Confidence Threshold	Accuracy	<u>Precision</u>	Recall	F1 Score	Avg Latency (ms)	Standard Deviation (ms)
<u>YOLOv11n</u>	4%	87.9%	92.0%	83.9%	87.8%	68.52	9.12
<u>YOLOv11s</u>	3%	88.3%	90.7%	86.3%	88.4	122.67	26.71





Testing Results

- On-device benchmark (Jetson Orin Nano):
 - YOLOv11n averaged 39.28 ms latency
 - ~25x faster than the 1 s requirement
- 30/30 successful detections on the edge device with stable runtime
- Live camera integration validated:
 - 29 fps effective throughput, 0 dropped frames
- End-to-end pipeline verified:
 - capture → inference → event → secure logging → dashboard
- YOLOv11n selected as final deployment model
 - lowest latency, lowest variability, smallest footprint
- Hardware constraint identified and mitigated:
 - memory limit addressed via swap provisioning

```
=====
EdgeSafe - Edge Platform Benchmark Results
=====
Timestamp:      2026-05-16 22:50:24.065792
Platform:       Linux-5.15.148-tegra-aarch64-with-glibc2.35
Python:         3.10.12
PyTorch:        2.8.0
CUDA enabled:   True
GPU:            Orin
Test image:     debug_detection.jpg
Runs/model:     50 (warmup 5)

[YOLOv11n]
Model:          ../runs/detect/yolov11n_firearm_v1/weights/best.pt
Classes (1):    {0: 'Guns'}
Conf threshold: 0.04
Detections:     50/50
Avg latency:    39.01 ms
Min / Max:      38.55 / 39.80 ms
Std deviation:  0.22 ms
Last detection: Guns:78.4%
SMART Obj #1 (<=1000ms): PASS

[YOLOv11s]      jessy Andujar, yesterday • Add edge inference verification
Model:          ../runs/detect/yolov11s_firearm_v12/weights/best.pt
Classes (1):    {0: 'Guns'}
Conf threshold: 0.03
Detections:     50/50
Avg latency:    44.20 ms
Min / Max:      43.66 / 49.40 ms
Std deviation:  1.21 ms
Last detection: Guns:82.9%
SMART Obj #1 (<=1000ms): PASS
```

```
(edgesafe-env) jessy@jessy-desktop:~/Desktop/Capstone-EdgeSafe
/mi$ python3 camera_capture.py --test
[test] Capturando 10s para medir FPS real...
[camera] Abierta /dev/video0 -> 1280x720 @ 30fps

[test] Resultados
      Frames OK:      290
      Frames fallidos: 0
      Tiempo:         10.0s
      FPS real:       29.0
      PASS - streaming continuo estable, 0 frames perdidos
(edgesafe-env) jessy@jessy-desktop:~/Desktop/Capstone-EdgeSafe
```



Conclusion





Conclusions

- EdgeSafe demonstrates that responsible, real-time firearm detection is feasible on low-cost edge hardware
- All core SMART objectives met: accuracy $\geq 80\%$, precision $\geq 90\%$, latency ≤ 1 s, secure auditable logging
- Edge-first design delivers privacy, low latency, and offline resilience: addressing the key gap in cloud-based solutions
- Live-camera stretch goal was achieved, exceeding the original proof-of-concept scope
- Limitation: single-semester proof-of-concept;
- Future work: extended reliability testing, multi-camera support, and full production deployment



Demo



Demo



Q&A: 5 mins